

50 MCQs with Explanation

Easy (10 MCQs)

1.

How do you start the OpenDaylight controller?

- a) Run the Karaf container
- b) Start a physical switch
- c) Open a web browser only
- d) Use FTP to upload files

Answer: a

Explanation: OpenDaylight runs inside the Karaf OSGi container.

2.

What is the purpose of the OpenFlow plugin in ODL?

- a) Communicate with OpenFlow switches
- b) Manage wireless devices
- c) Provide a web interface
- d) Store configuration files

Answer: a

Explanation: The plugin enables ODL to control OpenFlow-enabled switches.

3.

What does Mininet emulate?

- a) Virtual SDN networks with hosts and switches
- b) Physical hardware
- c) Email servers
- d) None of the above

Answer: a

Explanation: Mininet creates a virtual network for SDN testing.

4.

Which component in ODL manages data modeling?

- a) MD-SAL
- b) CLI
- c) OpenFlow plugin
- d) L2Switch app

Answer: a

Explanation: MD-SAL uses YANG models to manage network data.

5.

What does the internal datastore in ODL store?

- a) Network configuration and operational state
- b) Only user files
- c) Email logs
- d) None of the above

Answer: a

Explanation: The datastore stores configuration and current network state.

6.

What language does MD-SAL use to define data models?

- a) YANG
- b) XML only
- c) JavaScript
- d) Python

Answer: a

Explanation: YANG is the data modeling language used by MD-SAL.

7.

Which ODL application provides basic Layer 2 switching?

- a) L2Switch
- b) Firewall
- c) DHCP server
- d) None of the above

Answer: a

Explanation: L2Switch learns MACs and forwards packets like a normal switch.

8.

OpenDaylight clustering improves:

- a) Scalability and fault tolerance
- b) Only speed of packet forwarding
- c) User interface quality
- d) None of the above

Answer: a

Explanation: Clustering ensures high availability and load distribution.

9.

Open vSwitch is:

- a) A software OpenFlow switch
- b) A hardware router
- c) An email client
- d) None of the above

Answer: a

Explanation: OVS is software that supports OpenFlow.

10.

The southbound interface in ODL is responsible for:

- a) Communicating with network devices

- b) Managing user accounts
- c) Running web applications
- d) None of the above

Answer: a

Explanation: Southbound interfaces connect the controller to devices.

Medium (30 MCQs)

11.

Which command is used to install features in ODL Karaf?

- a) feature:install
- b) install:feature
- c) start:feature
- d) load:plugin

Answer: a

Explanation: Karaf uses feature:install to add new modules.

12.

How does ODL maintain consistency in clustered mode?

- a) Using distributed datastores
- b) Manual syncing by admin
- c) External databases only
- d) None of the above

Answer: a

Explanation: Distributed datastores keep all nodes in sync.

13.

Which datastore reflects the actual state of the network devices?

- a) Operational datastore
- b) Config datastore
- c) Candidate datastore
- d) None of the above

Answer: a

Explanation: Operational datastore reflects real-time device status.

14.

What is the default SDN protocol supported by ODL?

- a) OpenFlow
- b) BGP
- c) OSPF
- d) RIP

Answer: a

Explanation: OpenFlow is the main protocol ODL supports for SDN control.

15.

Which of these is NOT true about Mininet?

- a) It can emulate networks with virtual hosts and switches
- b) It requires multiple physical machines
- c) It supports OpenFlow switches like OVS
- d) It is used for SDN testing

Answer: b

Explanation: Mininet runs on a single machine, emulating multiple hosts.

16.

MD-SAL supports which type of data stores?

- a) Config, operational, candidate

- b) Only config
- c) Only operational
- d) None of the above

Answer: a

Explanation: MD-SAL supports all three datastore types.

17.

Which of the following best describes L2Switch?

- a) A simple application that learns MAC addresses and forwards packets
- b) A DHCP server
- c) An authentication service
- d) A web server

Answer: a

Explanation: L2Switch acts like a traditional Layer 2 switch.

18.

Which ODL feature manages service lifecycle and module dependencies?

- a) Karaf container
- b) OpenFlow plugin
- c) Mininet
- d) None of the above

Answer: a

Explanation: Karaf manages modules and their dependencies.

19.

What is the function of the OpenFlow plugin in ODL?

- a) To manage OpenFlow sessions with switches
- b) To configure wireless access points
- c) To manage email services
- d) None of the above

Answer: a

Explanation: It manages communication and flow installation on switches.

20.

Which protocol does Open vSwitch primarily use for controller communication?

- a) OpenFlow
- b) BGP
- c) DHCP
- d) FTP

Answer: a

Explanation: OVS supports OpenFlow for SDN control.

21.

How does OpenDaylight support multi-tenancy?

- a) Using isolated network slices via MD-SAL models
- b) By running multiple OS instances
- c) Through hardware virtualization only
- d) None of the above

Answer: a

Explanation: MD-SAL allows isolation and segmentation for tenants.

22.

What is the purpose of the candidate datastore in ODL?

- a) It acts as a staging area for config changes before committing
- b) Stores logs only
- c) Monitors user activity
- d) None of the above

Answer: a

Explanation: Candidate datastore is used for transactional config changes.

23.

Which ODL project provides a GUI for network visualization?

- a) DLUX
- b) Mininet
- c) Karaf
- d) L2Switch

Answer: a

Explanation: DLUX provides a web GUI for ODL.

24.

What role does the southbound plugin play?

- a) Translates southbound protocols to ODL internal model
- b) Exposes APIs to applications
- c) Provides CLI commands
- d) None of the above

Answer: a

Explanation: It acts as a translator between devices and MD-SAL.

25.

Which component in ODL handles flow rule installation?

- a) OpenFlow plugin
- b) Karaf
- c) L2Switch app
- d) Mininet

Answer: a

Explanation: OpenFlow plugin sends flow mod messages to switches.

26.

What is OpenDaylight's primary programming language?

- a) Java
- b) Python
- c) C++
- d) Ruby

Answer: a

Explanation: ODL is primarily written in Java.

27.

Which statement about Open vSwitch is correct?

- a) It can be controlled by ODL using OpenFlow
- b) It is a physical switch only
- c) It doesn't support SDN protocols
- d) None of the above

Answer: a

Explanation: OVS is software but controlled using OpenFlow.

28.

How does clustering in ODL improve reliability?

- a) Failover between nodes if one fails
- b) Increasing bandwidth only
- c) Providing static routing only
- d) None of the above

Answer: a

Explanation: Clustering provides redundancy and failover.

29.

What is the default port number for the ODL REST API?

- a) 8181
- b) 80
- c) 443

d) 22

Answer: a

Explanation: ODL REST API listens on port 8181 by default.

30.

Which component manages northbound APIs?

a) MD-SAL

b) OpenFlow plugin

c) Karaf

d) L2Switch app

Answer: a

Explanation: MD-SAL exposes northbound APIs to applications.

Hard (10 MCQs)

31.

How does MD-SAL achieve abstraction across different device types?

a) By using YANG models and southbound plugins

b) Through direct device drivers only

c) Using static configuration files only

d) None of the above

Answer: a

Explanation: YANG models define a common data model abstracting device details.

32.

What is a major challenge when clustering ODL controllers?

- a) Maintaining data consistency across nodes
- b) Setting up physical cables
- c) Writing Java code manually for failover
- d) None of the above

Answer: a

Explanation: Distributed state synchronization is complex in clustering.

33.

What is the role of the OpenFlow “Flow Table” in OVS controlled by ODL?

- a) It stores flow entries installed by the controller
- b) It stores user credentials
- c) It manages DNS records
- d) None of the above

Answer: a

Explanation: Flow tables contain forwarding rules from the controller.

34.

Which OpenDaylight project provides a virtual switch for SDN experiments?

- a) Open vSwitch (OVS)
- b) Karaf
- c) DLUX
- d) Mininet

Answer: a

Explanation: OVS is the virtual switch project.

35.

Which ODL datastore is transactional and supports rollback?

- a) Candidate datastore
- b) Operational datastore
- c) Config datastore

d) None of the above

Answer: a

Explanation: Candidate datastore allows transactional config changes.

36.

Which of these protocols besides OpenFlow can ODL support on southbound?

a) NETCONF

b) SMTP

c) FTP

d) HTTP

Answer: a

Explanation: NETCONF is used for device management and config.

37.

How does the L2Switch app handle unknown destination MAC addresses?

a) It floods the packet to all ports except source

b) Drops the packet

c) Sends packet to controller only

d) None of the above

Answer: a

Explanation: Unknown MACs are flooded to discover destination.

38.

Which mechanism in ODL ensures data consistency across cluster nodes?

a) Distributed data store replication

b) Manual syncing by admins

c) External databases only

d) None of the above

Answer: a

Explanation: Replication keeps data consistent across cluster members.

39.

What is the function of the OpenDaylight Karaf container?

- a) Module lifecycle management and OSGi runtime
- b) Packet forwarding
- c) Web hosting
- d) None of the above

Answer: a

Explanation: Karaf runs ODL as modular OSGi bundles.

40.

How does Mininet integrate with OpenDaylight?

- a) By connecting virtual switches (OVS) to ODL controller via OpenFlow
- b) By running ODL inside Mininet only
- c) By replacing ODL controller
- d) None of the above

Answer: a

Explanation: Mininet virtual switches communicate with ODL.

41.

Which of the following is NOT true about Open vSwitch?

- a) Supports OpenFlow 1.3 and above
- b) Is hardware only
- c) Runs on Linux
- d) Can be controlled remotely

Answer: b

Explanation: OVS is software-based, not hardware.

42.

What is the key advantage of the MD-SAL architecture?

- a) Device agnostic programming and consistent API
- b) No network programmability
- c) Only supports one vendor devices
- d) None of the above

Answer: a

Explanation: MD-SAL abstracts device details with YANG models.

43.

Which of the following ODL features enables service orchestration?

- a) Network Intent APIs
- b) CLI only
- c) OpenFlow plugin only
- d) None of the above

Answer: a

Explanation: Intent APIs simplify application-level programming.

44.

What is the primary communication method between ODL and OVS?

- a) OpenFlow
- b) SNMP
- c) HTTP
- d) SSH

Answer: a

Explanation: OpenFlow controls the switch.

45.

Which ODL component exposes northbound REST APIs?

- a) MD-SAL
- b) Karaf container
- c) L2Switch app

d) None of the above

Answer: a

Explanation: MD-SAL exposes northbound APIs for apps and services.

46.

How can you verify if the OpenFlow plugin is active in ODL?

a) Use feature:list | grep openflow in Karaf shell

b) Check hardware switches only

c) Use external monitoring tools only

d) None of the above

Answer: a

Explanation: Karaf commands show installed features.

47.

What type of flow entry action is typically used by L2Switch to forward packets?

a) Output to a specific port

b) Drop packet

c) Send to controller only

d) None of the above

Answer: a

Explanation: L2Switch installs output actions based on MAC learning.

48.

How does ODL handle failures in a clustered environment?

a) Leader election and failover mechanisms

b) Manual restart only

c) Hardware reset required

d) None of the above

Answer: a

Explanation: Cluster nodes elect leaders and failover automatically.

49.

Which tool can be used to create custom network topologies in Mininet?

- a) Python scripts
- b) HTML editors
- c) Command prompt only
- d) None of the above

Answer: a

Explanation: Mininet is programmable using Python.

50.

How do you add a new southbound protocol to OpenDaylight?

- a) Develop a southbound plugin integrating with MD-SAL
- b) Change hardware switch firmware
- c) Use CLI commands only
- d) None of the above

Answer: a

Explanation: Southbound protocols are integrated via plugins.